

Problem Statement

Nowadays, supermarkets and retail stores are facing a major problem because of increasing theft activities. Many customers may try to steal products by hiding items inside bags or clothes, changing barcodes, skipping billing counters, or leaving the store without paying. These activities cause huge financial losses to supermarket owners every year.

Most supermarkets use CCTV cameras for security, but these systems still depend mainly on human monitoring. Security guards or staff members have to continuously watch multiple camera screens for long hours, which is very difficult and tiring. In crowded environments, it becomes almost impossible to monitor every customer properly. Due to this, many suspicious activities are missed, and theft is often discovered only after the loss has already happened.

Another issue with existing surveillance systems is the large number of false alarms. Sometimes normal customer behavior is mistakenly treated as suspicious, making it harder for security staff to identify real threats quickly.

To solve these problems, there is a need for a smart and intelligent system that can automatically monitor customer activities in real time, identify suspicious behavior accurately, and immediately alert security staff with proper video evidence before the theft takes place.

Proposed Solution

To overcome the problems faced by traditional security systems, the proposed project introduces an **AI-Based Smart Theft Detection System** using technologies such as Machine Learning, Deep Learning, Computer Vision, and Video Analytics.

The system continuously analyzes live video footage captured from CCTV cameras installed inside the supermarket. Instead of relying completely on human observation, the AI system automatically studies customer movements and behavior patterns to identify suspicious activities.

The proposed system can detect activities such as:

- Hiding products inside bags or clothing
- Skipping billing counters without payment
- Removing security tags from products
- Staying in restricted areas for a long time
- Unusual or suspicious movement patterns
- Exiting the store with unpaid items

Whenever suspicious activity is detected, the system immediately sends alerts to the security staff along with captured images or video clips as evidence. This helps security personnel take quick action before the theft is completed.

The system is specially designed to work effectively even in crowded supermarket environments while reducing false alarms caused by normal shopping behavior. This improves overall security, reduces financial losses, and makes surveillance smarter and more efficient.